

SciAtlas: A Large-Scale Knowledge Graph for Automated Scientific Research

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ABSTRACT

The exponential growth of global academic output has confronted researchers and AI agents with an unprecedented “information explosion,” where fragmented and unstructured knowledge organization impedes deep interdisciplinary integration. Current academic retrieval tools predominantly rely on superficial keyword matching or vector-space semantic retrieval, which lack the topological reasoning capabilities required to navigate complex logical connections. Agentic deep-research-based frameworks are often prone to logical hallucinations and consuming high inference costs. To bridge this gap, in this report, we introduce SciAtlas, a large-scale, multi-disciplinary, heterogeneous academic resource knowledge graph designed as a panoramic scientific evolution network. By integrating over 43M papers from 26 disciplines, and a total of 157M entities and 3B triplets, SciAtlas provides a structured topological cognitive substrate that dismantles disciplinary barriers and furnishes AI agents with a global perspective. Furthermore, we develop a neuro-symbolic retrieval algorithm featuring tri-path collaborative recall and graph reranking, achieving a seamless transition from simple semantic matching to deterministic association discovery. We also present key application directions of SciAtlas, including literature review, automated research trend synthesis, idea positioning, and academic trajectory exploration, to demonstrate that SciAtlas can serve as an effective “cognitive map” to empower the full loop of automated scientific research while significantly reducing reasoning costs. We have released the interfaces for KG retrieval and various downstream tasks in our GitHub repo.

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llmXive follow-up. A separate llmXive study building on this work: [PROJ-854-llmxive-follow-up-extending-sciatlas-a-l](#).

SCIATLAS: A Large-Scale Knowledge Graph for Automated Scientific Research

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🌐 <https://github.com/zjunlp/SciAtlas>

Abstract The exponential growth of global academic output has confronted researchers and AI agents with an unprecedented “information explosion,” where fragmented and unstructured knowledge organization impedes deep interdisciplinary integration. Current academic retrieval tools predominantly rely on superficial keyword matching or vector-space semantic retrieval, which lack the topological reasoning capabilities required to navigate complex logical connections. Agentic deep-research-based frameworks are often prone to logical hallucinations and consuming high inference costs. To bridge this gap, in this report, we introduce **SCIATLAS**, a large-scale, multi-disciplinary, heterogeneous academic resource knowledge graph designed as a panoramic scientific evolution network. By integrating **over 43M papers from 26 disciplines, and a total of 157M entities and 3B triplets**, SCIATLAS provides a structured topological cognitive substrate that dismantles disciplinary barriers and furnishes AI agents with a global perspective. Furthermore, we develop a neuro-symbolic retrieval algorithm featuring tri-path collaborative recall and graph reranking, achieving a seamless transition from simple semantic matching to deterministic association discovery. We also present key application directions of SCIATLAS, including literature review, automated research trend synthesis, idea positioning, and academic trajectory exploration, to demonstrate that SCIATLAS can serve as an effective “cognitive map” to empower the full loop of automated scientific research while significantly reducing reasoning costs. We have released the interfaces for KG retrieval and various downstream tasks in our GitHub repo.

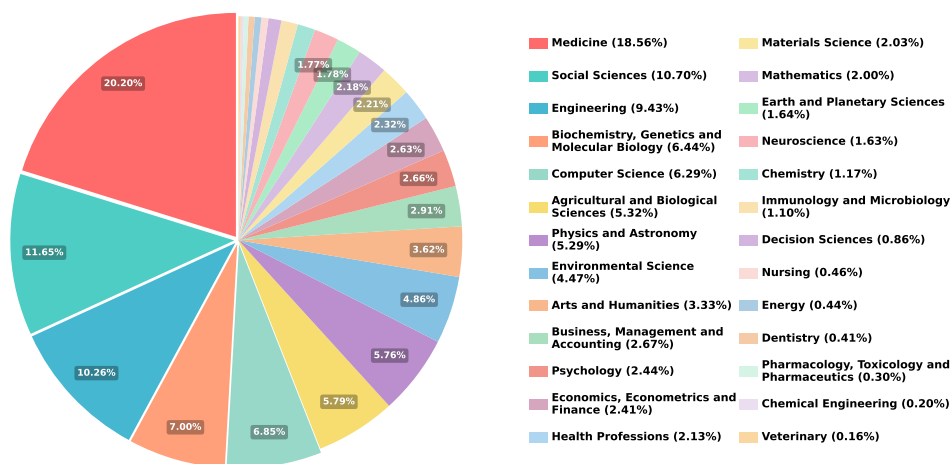


Figure 1 | Discipline Distribution in SCIATLAS. SCIATLAS is a large-scale scientific knowledge graph containing 26 disciplines with over 43M academic papers and other heterogeneous entities.

1. Introduction

Automated Scientific Research driven by Large Language Models (LLMs) has emerged as one of the most cutting-edge focal points in the field of artificial intelligence [Chen et al., 2025c, Lu et al., 2024, Shao et al., 2025]. With the exponential growth of global academic output, researchers and AI agents are jointly confronted with an unprecedented “information explosion” challenge. Precise literature retrieval and effective knowledge integration not only constitute the logical starting point of the research loop but also serve as the core cornerstone determining the success of subsequent innovation generation and experimental design [Moussa et al., 2025, Qiao et al., 2026, Tang et al., 2025, Zhang et al., 2026]. However, current academic retrieval tools are generally plagued by two major issues.

First is the *organizational form of academic knowledge*. Currently, vast amounts of research achievements are scattered across the internet in unstructured textual formats, lacking unified organizational paradigms and association mechanisms. This “knowledge island” phenomenon not only impedes deep interdisciplinary integration but also renders the intrinsic logical connections between entities latent and inaccessible. Novice researchers and AI agents struggle to transcend disciplinary barriers to perceive the global topological structure of scientific knowledge, resulting in cognitive dimensional deficits when addressing cutting-edge interdisciplinary topics [Ding et al., 2025].

Second is the *retrieval paradigm of academic knowledge*. Existing retrieval tools primarily rely on superficial keyword matching or vector-space-based semantic retrieval [Moussa et al., 2025, Ou et al., 2025, Qiao et al., 2026, Tang et al., 2025], both of which are essentially flattened feature comparisons and cannot support genuine topological reasoning. Some deep-research-based agentic frameworks attempt to compensate for the deficiency of structured information through iterative knowledge search and integration [Ahmad and Kim, 2026, Asai et al., 2024, Ju et al., 2025, Qian et al., 2026]. However, this approach not only incurs high computational costs and response latency but also, due to the absence of deterministic cognitive maps as anchors for LLMs, renders them highly susceptible to logical hallucinations within complex exploratory trajectories.

We introduce **SCIATLAS**¹, a large-scale, multi-disciplinary, heterogeneous academic resource knowledge graph designed to provide a topological cognitive substrate for accelerating scientific discovery. In terms of organizational structure, SCIATLAS features a sophisticated schema (see Fig.2) encompassing 9 categories of entity nodes, including papers, authors, institutions, keywords, research fields, etc. Each node type is endowed with comprehensive attribute information (e.g., paper abstracts and PDF URLs, author citations), as well as 12 categories of relational edges, including citations, authorship, co-authorship, keyword co-occurrence, etc. This organizational paradigm weaves fragmented knowledge into a self-explanatory, panoramic scientific evolution network. Such structured formalization can dismantle disciplinary barriers, elevating scientific research into an interconnected logical topology that furnishes AI agents with a global cognitive perspective for observing scientific advancement.

Building on SCIATLAS, we develop a neuro-symbolic retrieval algorithm that achieves the transition from semantic matching to topological reasoning. By integrating lexical matching, vector retrieval, and well-developed graph propagation algorithms [Tong et al., 2006], we establish a tri-path collaborative recall and graph reranking mechanism, which enables deep fusion of the semantic relevance of papers, graph topological support, and importance metrics based on global citations, thereby providing deterministic deep association discovery without requiring frequent iterations of LLMs and high reasoning costs. Furthermore, we propose several potential downstream application directions of SCIATLAS for automated scientific research, including literature review, differentiated positioning and similarity detection of research ideas, idea generation, automated research trend predicting,

¹This project is part of the SciGraph project (<http://scigraph.openkg.cn/>) under SciGraph-Scholar.

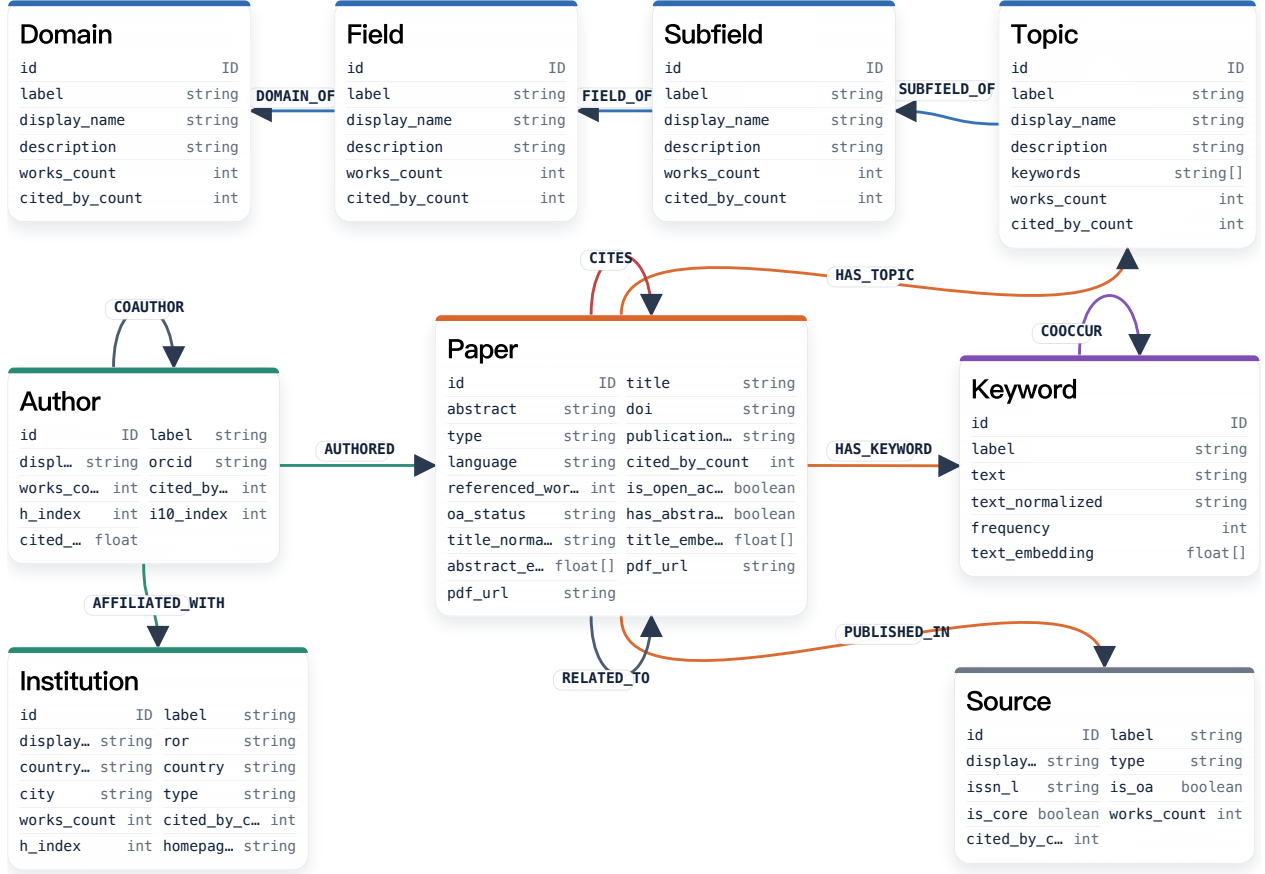


Figure 2 | Schema of SCIATLAS. By integrating 9 kinds of entity nodes and 12 kinds of relational edges, SCIATLAS provides a structured topological cognitive substrate that dismantles disciplinary barriers and furnishes AI agents with a global perspective. The complete schema (including entities, relations, attributions) of SCIATLAS can be found in Appx.A.

retrieval of highly relevant academic authors, and academic trajectory exploration for researchers.

Our main contributions are as follows:

- We introduce SCIATLAS, a large-scale, multi-disciplinary knowledge graph that organizes fragmented academic resources into a structured logical topology. It serves as a comprehensive, panoramic scientific network that provides AI agents with a global cognitive perspective.
- We develop an efficient neuro-symbolic retrieval algorithm featuring tri-path collaborative recall and graph reranking, achieving the transition from surface-level semantic matching to deterministic topological reasoning.
- We propose application directions for SCIATLAS, including research trend synthesis, idea positioning, and academic trajectory exploration, etc. These applications demonstrate SCIATLAS’s capability as a “cognitive map” to empower the entire loop of automated scientific research.

2. SCIATLAS

Table 1 | Statistics of SciATLAS. SciMap comprises a total node count reaching tens of millions, with the aggregate number of edges scaling to billions.

Entity (Total: 157M)				Relation (Total: 3B)			
Type	Num	Type	Num	Type	Num	Type	Num
Paper	43.30M	Author	109.70M	(Paper, CITES, Paper)	213.88M	(Paper, HAS_KEYWORD, Keyword)	101.38M
Keyword	3.76M	Institution	0.12M	(Paper, HAS_TOPIC, Topic)	105.89M	(Author, AFFILIATED_WITH, Instit)	195.94M
Topic	4.52K	Subfield	252	(Author, AUTHORED, Paper)	149.00M	(Author, COAUTHOR, Author)	2.06B
Field	26			(Keyword, COOCCUR, Keyword)	60.37M	(Field, DOMAIN_OF, Domain)	26
Source	0.28M			(Subfield, FIELD_OF, Field)	252	(Paper, RELATED_TO, Paper)	68.38M
Domain	4			(Topic, SUBFILED_OF, Subfield)	4.52K	(Paper, PUBLISH_IN, Source)	40.90M

2.1. Overview of SciATLAS

Schema. In Fig.2, we present the complete schema of SciATLAS. SciATLAS is constructed with academic literature as its core, encompassing entities such as Author, Institution, Keyword, Source, Topic, Field, Subfield, and Domain centered around the Paper entity. With the help of these hybrid entities, the papers are organized directly or indirectly in four levels:

- **Semantic level.** The citation relationship (CITES) and relevance relationship (RELATED_TO) establish direct semantic connections between papers.
- **Conceptual level.** Each paper is associated with its most salient keywords, and the COOCCUR relationships among keywords within papers indirectly link papers at the conceptual level.
- **Direction level.** Different domains, fields, subfields, and topics organize papers into hierarchical structures at the disciplinary and research direction levels.
- **Social level.** COAUTHOR relationships among authors and AUTHORED relationships between authors and papers, together with the AFFILIATED_WITH relationships between authors and institutions, form indirect relationships between papers at the social organizational level.

These multi-level organizational structures constitute a complex paper relationship network, providing a robust structural foundation for deep retrieval and reasoning over SciATLAS.

Statistics. SciATLAS covers 26 academic disciplines (see Fig.1) with a total of 43.30 million papers. Medicine holds the largest share (18.56%), followed by Social Sciences (10.70%), Engineering (9.43%), Biochemistry, Genetics and Molecular Biology (6.44%), and Computer Science (6.29%). The five disciplines above collectively account for 51.43% of the total paper volume, reflecting the concentration of core disciplines. The remaining fields range from Arts and Humanities (3.33%) to Veterinary (0.16%), ensuring broad disciplinary representation. In terms of scale, in Tab.1, SciATLAS contains 109.70 million authors, 3.76 million keywords, and 0.12 million institutions, connected by billions of relational edges across 11 relationship types. This combination of comprehensive disciplinary coverage and massive entity volume positions SciATLAS as a large-scale, multi-disciplinary knowledge graph for topological scientific search.

2.2. SciATLAS Construction

The primary data source for our knowledge graph is from OpenAlex², a fully open-source library of scholarly resources encompassing over 480 million academic publications. Each paper contains rich metadata, including authors, abstracts, institutions, publication dates, venues, references, citation counts, topics, open-access status, PDF URL, etc. Building upon this foundation, we construct our knowledge graph through the following primary steps:

²<https://openalex.org/>.

Data Restructuring and Filtering. First, we extract different entity types from OpenAlex and preserve only key attributes for each entity. Subsequently, since OpenAlex data is also sourced from the internet and contains substantial noise, we normalize and deduplicate the names of various entities (e.g., paper titles, institution names) after standardization. Notably, we do not deduplicate authors due to the prevalence of name duplication and ambiguity. We also discard entities lacking critical attributes (e.g., paper PDF URLs). We then filter out non-English papers and papers with very short abstracts to ensure high-quality. Next, we establish edges based on the inter-entity information stored within each entity (e.g., authors and references contained in papers). Since OpenAlex assigns a unique ID to each entity, we directly utilize these IDs to match corresponding entities and construct relationships.

Keyword Extraction. Although OpenAlex includes a Concept entity type as the core concept of papers, it is excessively sparse (only 65K entries, far fewer than the 480M paper corpus) and more critically, these concepts remain at a macroscopic and superficial level (e.g., “artificial intelligence”), failing to genuinely represent the core concepts and terms within individual papers. These limitations are insufficient for complex academic relational reasoning in KG, motivating us to construct denser and truly useful keywords. Specifically, we employ a lightweight open-source LLM (Qwen3-30B-A3B-Instruct-2507 [Yang et al., 2025]) as an extractor to identify keywords from paper abstracts. Recognizing that many contemporary papers tend to emphasize narrative packaging, which often obscures their academic essence, and the same concept may be expressed differently across distinct domains, we deliberately instruct the LLM to avoid paper-specific terminology or system names, as well as highly customized or marketing-style expressions. Instead, we prioritize those fundamental phrases that are reusable across numerous papers. For each paper, we extract 3-8 core keywords to constitute the Keyword entity. The LLM will also assign an *importance score* to each keyword, which serves as the attribute for the HAS_KEYWORD edge. Please see Appx.B.1 for the detailed prompt of keyword extraction. To capture associations among keywords, we establish COOCUR relations between keywords appearing in the same paper, with *co-occurrence frequency* serving as edge weights to indicate the strength of association between keywords.

Examples of Good and Bad Keywords

Good Keywords: protein structure prediction, idea evaluation, wireless communication, energy optimization, fault detection, monte carlo simulation

Bad Keywords: hierarchical dual-path adaptive learning framework, multi-stage cross-modal feature fusion architecture, novel high-performance prototype system, AlphaEvolve

Semantic Embedding. To support hybrid and efficient KG retrieval, we incorporate pre-computed semantic vectors into SCIATLAS in addition to plain text. Specifically, we select the three most semantically rich fields: paper title, paper abstract, and keyword. We first normalize each field (format and case), then employ bge-large-en-v1.5 [Xiao et al., 2024] as the embedding model. The semantic vectors derived from the titles and abstracts are integrated as paper attributes, while those derived from the keywords are incorporated as keyword attributes.

Finally, we organize all entities, attributes, and edges together and deploy SCIATLAS using Neo4j³.

2.3. SCIATLAS Update

To accommodate rapid knowledge iteration, we propose several approaches for SciMap updates:

³<https://neo4j.com/>.

Using with Online Resources. OpenAlex provides daily-updated API endpoints⁴ supporting daily updates for entities such as papers, authors, and institutions. Users can retrieve information for desired papers directly through the API, follow the pipeline described in §2.2 to extract keywords, compute semantic embeddings, and extract inter-entity relationships aligned with the SCIATLAS schema, and finally import them into the database via Neo4j Cypher language. Although OpenAlex encompasses the vast majority of literature available on the internet, rare cases of absent papers may occur. For such scenarios, we recommend GROBID⁵, a very lightweight information extraction tool specifically designed for technical and scientific publications, which can rapidly extract metadata, including titles, authors, abstracts, and references, from paper’s PDF file, serving as an efficient alternative to the OpenAlex API. We will open our KG construction code to support the evolution.

Periodic Update. OpenAlex compiles changefiles⁶ of the latest updates every two months compared to the previous version. Our team will periodically update our knowledge graph based on these releases. Users who have already deployed the system locally can also maintain their knowledge graph periodically. Our pipeline supports one-click import from OpenAlex downloaded files to SCIATLAS.

3. Neuro-Symbolic Retrieval

In this section, we introduce a neuro-symbolic retrieval algorithm featuring tri-path collaborative entity recall and achieve deep topological reasoning through graph traversal. It can also serve as a fundamental retrieval algorithm adaptable to various downstream tasks in §4.

3.1. Node Matching

Our retrieval system supports arbitrary query formats, including keywords, scientific questions, abstracts, idea texts, and even complete papers. Given a query q , we map it into KG nodes through three distinct ways.

Keyword Matching. We use an LLM to extract keywords from q and assign each keyword with an importance score, forming a keyword list $\mathcal{K} = \{(k_i, s_i^{\text{llm}})\}_{i=1}^m$, where k_i is the i -th extracted keyword with text normalization and $s_i^{\text{llm}} \in [0, 1]$ represents its normalized importance score. The maximum number of keywords extracted by the LLM is m . Then, we first perform exact text matching of k_i in the KG. For each matched keyword node g , we assign it an exact match score:

$$\text{score}_{\text{exact}}(k_i, g) = s_i^{\text{llm}}. \quad (1)$$

Second, we perform vector matching. After encoding each k_i into a semantic vector, we compute semantic similarity based on the pre-calculated keyword text embeddings in the KG. Nodes with similarity scores exceeding the threshold θ_{kw} (default to 0.7) are retained, with their scores as:

$$\text{score}_{\text{vec}}(k_i, g) = s_i^{\text{llm}} \cdot \text{sim}(k_i, g). \quad (2)$$

If multiple nodes surpass the threshold, we select only the top-3 nodes for each k_i . The same keyword node g may be matched by multiple input keywords or simultaneously by both exact and vector matching. We take the maximum of all its scores as the node’s final weight:

$$w_g^{kw} = \max_i \left(\mathbb{1}[k_i = g] \cdot s_i^{\text{llm}}, \mathbb{1}[\text{sim}(k_i, g) \geq \theta_{kw}] \cdot s_i^{\text{llm}} \text{sim}(k_i, g) \right) \quad (3)$$

The final set of keyword-matching nodes is denoted as $\mathcal{K}_{\text{seed}} = \{(g, w_g^{kw})\}$.

⁴<https://developers.openalex.org/api-reference/introduction>.

⁵<https://github.com/grobidOrg/grobid>.

⁶<https://developers.openalex.org/download/changefiles>.

Semantic Matching. We embed query q to obtain vector $\mathbf{e}_q$ (Here, if the input is an entire paper, we only extract its abstract for embedding.), which is then used to retrieve the top-60 papers from the KG based on title embeddings and abstract embeddings, respectively. We then employ a reranker (bge-reranker-large [Xiao et al., 2024]) to re-rank the retrieved papers, retaining the top-15 papers for title and abstract. Given a retrieved paper p , we define s_p^{title} and s_p^{abs} as its retrieval scores through title or abstract matching, and compute a weighted combination of the two scores:

$$s_p^{emb} = \frac{0.4 \cdot s_p^{title} + 0.6 \cdot s_p^{abs}}{0.4 \cdot \mathbb{1}[\exists s_p^{title}] + 0.6 \cdot \mathbb{1}[\exists s_p^{abs}]} . \quad (4)$$

Here, it is set to 0 if s_p^{title} or s_p^{abs} does not exist. The final candidate paper nodes from semantic matching are denoted as $\mathcal{P}^{emb} \{(p, s_p^{emb})\}$.

Title Matching. Since titles encapsulate the most critical information of papers and are highly beneficial for paper retrieval, we specifically perform title matching for queries q that contain titles. We use GROBID to extract all titles (including the paper’s title and its references’ titles) from the idea or paper and employ an LLM to assign a confidence score c_j to each title t_j . We retain the top-10 titles with the highest confidence scores and normalize them (removing non-alphabetic characters and converting to lowercase) to obtain the title set $\mathcal{T} = \{(t_j, c_j)\}_{j=1}^n$. We then perform text matching of titles in the KG. If an exact match is found, a matching score of $m(t_j, p) = 1.0$ is assigned; otherwise, we compute the fuzzy similarity between two titles based on the following formula:

$$m(t_j, p) = 0.65 \cdot \text{seq}(t_j, p) + 0.35 \cdot \text{token_overlap}(t_j, p), \quad (5)$$

where $\text{seq}(a, b)$ is based on the Longest Common Subsequence (LCS) of a and b , and token_overlap computes the Jaccard overlap ratio of the token sets of a and b . Candidates with similarity below θ_{title} (default to 0.88) are directly discarded. For paper p matched by title t_j , we assign it a score:

$$s_{j,p}^{title} = c_j \cdot m(t_j, p). \quad (6)$$

If the same paper is matched by multiple titles, we take the maximum score $s_p^{title} = \max_j s_{j,p}^{title}$. Each input title retains at most the top-5 papers, and all papers constitute $\mathcal{P}^{title} = \{(p, s_p^{title})\}$.

Node Merging. We obtain two candidate paper node sets through the semantic and title pathways. Then we need to merge them into $\mathcal{P}_{seed}$ and unify their weights. For each candidate paper p , we compute the dot product with vector $\mathbf{e}_q$ and apply weighting according to the ratio specified in Eq.4:

$$\tilde{s}_p^{emb} = \text{combine}(\text{sim}_p^{title}, \text{sim}_p^{abs}), \quad \text{sim}_p^{title} = \mathbf{e}_q^\top \mathbf{e}_p^{title}, \quad \text{sim}_p^{abs} = \mathbf{e}_q^\top \mathbf{e}_p^{abs}. \quad (7)$$

We then perform MinMax normalization:

$$\tilde{s}_p^{emb} = \text{MinMaxNorm}(\tilde{s}_p^{emb}), \quad \tilde{s}_p^{title} = \text{MinMaxNorm}(s_p^{title}), \quad \text{MinMaxNorm}(x_p) = \frac{x_p - x_{min}}{x_{max} - x_{min}} \quad (8)$$

Finally, we define the unified paper weight:

$$s_p^{pre} = \lambda_{emb} \tilde{s}_p^{emb} + \lambda_{title} \tilde{s}_p^{title} + b_p^{pre}, \quad b_p^{pre} = \begin{cases} 0.35, & \text{exact title hit} \\ 0.10, & \text{fuzzy title hit} \\ 0, & \text{otherwise} \end{cases} \quad (9)$$

where b_p^{pre} denotes the title bonus, and λ_{emb} (default to 0.3) and λ_{title} (default to 0.8) represent the importance weights for semantic and title pathways, respectively.

3.2. Weight Setting

Taking $\mathcal{K}_{seed}$ and $\mathcal{P}_{seed}$ as starting points, we perform a 2-hop subgraph propagation, where all edges are treated as undirected during the propagation process. To prevent subgraph explosion, we select at most 500 nodes per hop for each entity type. For each paper p in the local subgraph, we compute its importance based on its citation count c_p . Let C denote total citation counts for all papers in the subgraph. The paper's importance is defined as:

$$\text{imp}(p) = \min \left(1, \frac{\log(1 + c_p)}{\log(1 + \max(1, C))} \right). \quad (10)$$

Here, the importance can be tailored to the downstream task: if the task emphasizes paper quality, it can be computed according to Eq.10; if the focus is solely on relevance, all papers can be forced to $\text{imp}(p) = 1$. For each seed paper p , we define its unnormalized weight as:

$$w_p^{seed} = s_p^{pre} \cdot (1 + \gamma \cdot \text{imp}(p)), \quad (11)$$

where γ is the control factor for importance (default to 0.5). For each seed keyword g , we define its unnormalized weight as $w_g^{seed} = w_g^{kw}$. We define the distribution $\mathbf{s}$ over all nodes in the graph as:

$$s_v = \begin{cases} \frac{w_v^{seed}}{Z}, & v \in S \\ 0, & v \notin S \end{cases}, \quad Z = \sum_{v \in S} w_v^{seed}, \quad S = \mathcal{P}_{seed} \cup \mathcal{K}_{seed}. \quad (12)$$

For an edge $e = (u, v, r)$ in the graph, we define its unnormalized weight $\omega(u, v)$ based on the edge type, as specified in Tab.2.

3.3. Random Walk with Restart

To more deeply explore the topological relationships between nodes and enable deep reasoning within the graph, we perform random walks on the graph based on seed nodes and edge weights. For any node u , let its neighbor set be $N(u)$. The transition probability from u to its neighbor v is defined as:

$$P(v | u) = \frac{\omega(u, v)}{\sum_{x \in N(u)} \omega(u, x)}. \quad (13)$$

Assuming the node score vector at iteration t is $\mathbf{r}^{(t)}$, we initialize $\mathbf{r}^{(0)} = \mathbf{s}$. For any node v , its score in the next iteration is:

$$r_v^{(t+1)} = \alpha s_v + (1 - \alpha) \sum_u r_u^{(t)} P(v | u), \quad (14)$$

where α denotes the restart probability. If a node u has no neighbors, we preserve its own mass by directly adding $(1 - \alpha)r_u^{(t)}$ back to u itself. The iteration terminates when:

$$\|\mathbf{r}^{(t+1)} - \mathbf{r}^{(t)}\|_1 < \varepsilon, \quad (15)$$

where $\varepsilon = 10^{-6}$, or when the maximum number of iterations $T_{\max} = 50$ is reached. The final graph score of node v is given by $r_v = r_v^{(t^*)}$, where t^* denotes the stopping iteration.

Table 2 | Definitions of Unnormalized Edge Weights ($\omega(u, v)$).

Edge Type	Weight Formula(s)	Parameter Description
HAS_KEYWORD	$\omega_{HK}(p, g) = \beta_{hk} \cdot \kappa(g) \cdot \text{rel}_{p,g}$ $\kappa(g) = \begin{cases} w_g^{kw}, & \text{if } g \text{ is a seed} \\ \epsilon_{kw}, & \text{otherwise} \end{cases}$	β_{hk} : Base weight for keyword association (default 1.20). $\text{rel}_{p,g}$: Importance score from (p, g) . $\kappa(g)$: Prior weight modulator for the keyword node. w_g^{kw} : Initial matching score for seed keywords. ϵ_{kw} : Smoothing factor for non-seed keywords (default 0.25).
CITES	$\omega_{CITES}(u, v) = \beta_{cite}$	β_{cite} : Base weight for paper citation relation (default 1.00).
RELATED_TO	$\omega_{RELATED}(u, v) = \beta_{rel}$	β_{rel} : Base weight for paper relatedness (default 0.90).
AUTHORED	$\omega_{AUTHORED}(u, v) = \beta_{auth}$	β_{auth} : Base weight for authorship relation (default 0.80).
COAUTHOR	$\omega_{COA}(u, v) = \beta_{coauth} \cdot \max(1, \phi(n_{uv}))$ $\phi(n_{uv}) = \min(c_{max}, \log(1 + n_{uv}))$	β_{coauth} : Base weight for co-authorship (default 0.60). n_{uv} : Co-authoring frequency. $\phi(\cdot)$: Frequency smoothing function. c_{max} : Logarithmic cap to prevent infinite weight magnification (default 2.0).
COOCCUR	$\omega_{COO}(u, v) = \beta_{cooc} \cdot \max(1, \phi(m_{uv}))$ $\phi(m_{uv}) = \min(c_{max}, \log(1 + m_{uv}))$	β_{cooc} : Base weight for keyword co-occurrence (default 0.60). m_{uv} : Co-occurrence frequency. $\phi(\cdot), c_{max}$: Same smoothing function and cap definition as COAUTHOR.

3.4. Final Ranking

Upon completing the graph propagation, the system derives a set of global node scores $\{r_v\}_{v \in V'}$ across the local subgraph. For the purpose of paper retrieval, we isolate the scores of paper nodes:

$$s_p^{graph} = r_p, \quad p \in V' \cap \text{Paper} \quad (16)$$

Crucially, this stage allows for the inclusion of newly discovered paper nodes that are not part of the initial $\mathcal{P}_{seed}$ sets but are reached during graph expansion.

To account for the academic impact of candidates within the retrieved context, we re-calculate the paper importance $\text{imp}_{final}(p)$ based on the citation distribution of the final candidate set. We utilize the logarithmic scaling defined in Eq.10, using total citations within the current pool to ensure a robust relative metric. To prevent the graph diffusion from over-promoting distant nodes, we introduce a graph support factor g_p , which acts as a gating mechanism based on the initial retrieval strength:

$$g_p = \max(0.25, \tilde{s}_p^{pre}), \quad (17)$$

where $\tilde{s}_p^{pre}$ is the MinMax-normalized pre-graph score s_p^{pre} in Eq.9. This ensures that while graph-discovered papers can achieve high ranks, those with zero initial semantic relevance must demonstrate exceptionally strong topological support to surpass primary candidates. We then normalize the graph score s_p^{graph} with MinMax to obtain $\tilde{s}_p^{graph}$.

The comprehensive final score s_p^{final} is defined as a weighted linear combination of three normalized components and a title-matching bonus:

$$s_p^{final} = \min \left(1, \lambda_{pre} \tilde{s}_p^{pre} + \lambda_{graph} \tilde{s}_p^{graph} g_p + \lambda_{imp} \text{imp}_{final}(p) \right), \quad (18)$$

where λ_{pre} (default to 0.35) is the initial relevance, λ_{graph} (default to 0.45) is the topological support from graph, and λ_{imp} (default to 0.20) is the citation importance. We finally return the **top-20 papers**, accompanied by a **detailed score breakdown** and **path-based explanations** to provide researchers with a transparent and deterministic “cognitive map” of the retrieval results. The entire retrieval process can be completed within 2 minutes, significantly shorter than LLM-based deep research frameworks, while still delivering high-relevance results with in-depth topological reasoning.

4. Downstream Application of SCIATLAS

Building upon SCIATLAS and our search algorithms, in this section, we propose several potential downstream applications of SCIATLAS to facilitate researchers’ scientific endeavors and accelerate automated scientific research. The detailed prompt used in this section can be found in Appx.B.2.

4.1. Literature Review

One of the most fundamental applications of scientific search is literature review, which essentially involves retrieving papers relevant to a given research direction and synthesizing a review report. We present a basic retrieval pipeline in §3, where users can customize retrieval based on their specific requirements for the retrieved literature. For instance, 1) if the retrieved papers are required to be published in top-tier conferences or journals, venue information can be incorporated into the importance score calculation of papers; 2) If author authority is emphasized, the citation count of authors can be reflected in the weight of AUTHORED edges; 3) If institutional authority is emphasized, corresponding weights can be assigned to AFFILIATED_WITH edges based on the reputation of institutions. Our algorithm provides flexible hyperparameter selection and functional adaptation to accommodate diverse retrieval focus requirements. We will progressively open configuration permissions for various hyperparameters of the retrieval algorithm to support user-customized retrieval. With the retrieved paper collection, it can be adapted to various LLM-based automated literature review methods [Liang et al., 2025, Wang et al., 2024b, Yan et al., 2025, Zhu et al., 2025].

4.2. Idea Grounding and Evaluation

Idea Grounding. By using the idea or paper as the query, we can retrieve a set of highly relevant papers from the KG and segment the full texts of these papers into finer-grained paragraphs. Subsequently, we employ an LLM to extract more refined queries or claims from the idea across multiple dimensions, including motivation, methodology, and experimental design, and use these refined queries to retrieve relevant paragraphs. Then, through LLM-based analysis, we identify the similarities and differences between the idea and the retrieved paragraphs. Through this entire pipeline, we can determine whether prior similar work exists for the idea, find evidence to support it, or identify its real innovative aspects. Since grounding may prioritize paper relevance, we can relax the emphasis on paper citations in Eq.11&18. We use [Qiao et al., 2026] as a running example in the following:

An Example of Idea Grounding

Target Idea or Paper: InnoEval: On Research Idea Evaluation as a Knowledge-Grounded, Multi-Perspective Reasoning Problem.

Target Query or Claim from the Idea: Mainstream approaches directly using LLM-as-a-Judge fossilize the models' inherent biases into de facto evaluation criteria, failing to emulate the deliberation among distinct perspectives needed for fair scientific evaluation.

Evidence Paper: Evaluating LLMs' Divergent Thinking Capabilities for Scientific Idea Generation with Minimal Context [Ruan et al., 2026].

Evidence Paragraph: Furthermore, we recognize a fundamental challenge in the reliability of LLM-as-a-Judge approaches. When evaluating scientific ideas containing concepts outside the judge models' knowledge boundaries, these models might misunderstand novel concepts and consequently misjudge their originality or feasibility. While our use of a dynamic panel of state-of-the-art judge models likely provides broader...

Matching Aspect: Limitations of LLM-as-a-Judge approaches.

Similar Point: Both identify a fundamental challenge or limitation with LLM-as-a-Judge approaches for evaluating scientific ideas. Both imply that LLM judges may produce unreliable or biased evaluations due to inherent model limitations.

Different Point: Target idea emphasizes failure to emulate multi-perspective deliberation for fair scientific evaluation; evidence focuses on bias and score caused by limited knowledge without addressing deliberation or multi-perspective aspects.

Idea Evaluation. With the grounding results, we can evaluate the idea by assessing its *novelty* based on the existence of prior similar work, its *feasibility* based on theoretical evidence, and its *soundness* based on the experimental designs of related studies. The focus of the grounding stage can be adjusted according to the criteria of downstream evaluation. This process can serve as a decision-making reference for human experts or be replaced by LLM-as-a-judge, becoming a critical evaluation signal for idea iteration in automated scientific discovery.

4.3. Idea Generation

We can use a research direction as the query, or an idea or a paper as the anchor, where retrieval in the KG functions as a knowledge collection process. The collected papers can be utilized for a literature review to identify gaps and propose new ideas, or to synthesize concepts from different domains and generate interdisciplinary ideas. Noting that the emergence of novel ideas typically stems from the fusion and refinement of two relatively distant concepts, we can relax the constraints on distant nodes in Eq. 17 during the search process to make the search more exploratory and the retrieved papers more diverse, thereby enhancing the novelty of generated ideas. Here we show an example by using "Knowledge Editing" as the query:

An Example of Idea Generation

Idea: Federated and Privacy-Preserving Knowledge Editing

Description : Design a knowledge editing framework suitable for a federated learning setting, where edits (e.g., corrections from user feedback) are computed locally on client devices and then aggregated to update a central model without exposing raw user data or the specific edits requested by individuals.

Novelty: All existing editing methods assume centralized access to the full model and edit dataset. This idea introduces the constraints of federated learning—data decentralization, privacy, and communication efficiency—to the knowledge editing problem, a combination not yet explored.

Significance: Enables large-scale, privacy-respecting model updates from distributed user interactions. This is crucial for applications like personal AI assistants on mobile devices, where users want to correct model errors without compromising their private data or queries.

Key References:

- [2024] Knowledge Editing on Black-box Large Language Models
- [2023] EasyEdit: An Easy-to-use Knowledge Editing Framework for Large Language Models
- [2025] Massive Editing for Large Language Models Based on Dynamic Weight Generation

4.4. Research Trend Predicting

For trend prediction in a specific research direction, the most critical aspect is understanding the current development status of that direction, which aligns with the objective of idea generation. The distinction lies in the fact that trend prediction emphasizes paper influence, as more impactful papers typically signify greater evolution in the research direction. Therefore, in this task, we can increase the importance weight of paper citations in Eq.11&18 during the search. Furthermore, to achieve a more comprehensive understanding of the field, we can relax the constraints on the number of papers retained during the search process and in the final results, making the retrieval outcomes more general. We can sort the retrieved papers chronologically and employ an LLM to summarize the developmental trajectory of the research direction, focusing on the discussion or limitation sections of papers to identify critical problems that need to be addressed and propose potential research directions for the future. Here is an example of research trend predicting by LLM:

An Example of Research Trend Predicting

Research Direction: Biologically plausible learning in spiking neural networks.

Stage Summary:

1. **2006-2014: Foundational Mechanisms.** Early exploration of biologically plausible learning rules for spiking networks, focusing on gradient estimation through dynamic perturbation, unsupervised learning via STDP, and basic cognitive function implementation.
2. **2015-2019: Cognitive and Sequence Learning.** Application of biologically plausible rules to more complex tasks, including goal-directed decision making, sequence learning, and pattern recognition, with growing emphasis on combining multiple plasticity mechanisms.
3. **2020-2022: Systematic Framework Development.** Concerted effort to develop learning frameworks as alternatives to backpropagation, addressing credit assignment problems and creating unified approaches that maintain biological plausibility while improving performance.
4. **2023-2025: Integration and Efficiency.** Focus on optimizing for efficiency and scalability, incorporating event-driven computation, exploring different neuron models, and developing novel mechanisms like bidirectional distillation for competitive performance.

Future Directions:

1. Development of fully event-driven, large-scale learning systems.
2. Integration of neuromodulation and attention mechanisms into learning frameworks.
3. Co-design of algorithms and neuromorphic hardware for optimal efficiency.
4. Exploration of meta-learning and continual learning in biologically plausible SNNs.
5. Bridging the gap between computational models and experimental neuroscience findings.

4.5. Related Author Retrieval

Given a research direction, retrieving relevant authors in that field can be as straightforward as simply replacing Eq.16 with:

$$s_a^{graph} = r_a, \quad a \in V' \cap \text{Author} \quad (19)$$

Subsequently, factors such as the authors' citation counts can serve as critical references for final ranking and filtering. During the retrieval process, to emphasize the contribution of authors to papers, we can adjust the weights of AUTHORED edges based on author order (e.g., increasing the transition probabilities for first and last authors relative to other authors) to retrieve the most relevant authors.

4.6. Researcher Background Review

Given an author, we can directly match his/her name to the graph node and collect all the author's published papers from the graph and summarize the author's academic background using an LLM. Since an author may simultaneously work on multiple research directions, we can first cluster the collected papers and have the LLM summarize within each cluster, then integrate them into a unified report. An LLM-generated researcher profile is shown below:

Researcher Background Review

*** is a prolific researcher with an academic trajectory spanning Natural Language Processing, Artificial Intelligence, and Large Language Models. His work demonstrates a strong emphasis on bridging symbolic knowledge (knowledge graphs) with statistical learning (large language models), particularly in the areas of information extraction, reasoning, and agentic systems. A significant recent pivot involves developing methods to understand, control, and align the internal mechanisms and behaviors of large-scale AI models, moving from application-focused to fundamental model analysis and intervention.

Research Trajectory:

1. Knowledge-Enhanced Language Models & Information Extraction (2018-2023)

Early and sustained focus on integrating structured knowledge (ontologies, knowledge graphs) into NLP models. This includes pioneering work on prompt-tuning for relation extraction (KnowPrompt), generative models for knowledge graph completion (GenKGC), and multimodal knowledge graphs. The goal is to make models more data-efficient, interpretable, and grounded in factual knowledge.

2. Reasoning, Planning, and Agentic AI Systems (2023-2026)

A major shift towards enabling LLMs to perform complex, multi-step reasoning and act as autonomous agents. Research focuses on augmenting agents with actionable knowledge (KnowAgent), refining their planning capabilities, and developing benchmarks for evaluation. This theme explores how to equip LLMs with procedural knowledge and reliable world models for task execution.

3. Model Analysis, Control, and Alignment (2023-2026)

A cutting-edge direction focused on diagnosing and steering the internal dynamics of LLMs. Work includes developing unified frameworks for understanding parameter dynamics from fine-tuning to activation steering, diagnosing truthfulness via consistency under perturbation, and predicting unintended behaviors from data. This represents foundational research into model interpretability and safety.

5. Limitations and Future Work

Our SCIATLAS is under continuous maintenance and updates. To further facilitate automated scientific discovery, we enumerate several important directions for future work.

CLI and Skills. Currently, our KG is primarily accessed through the Neo4j interface. Although we provide usage guidelines, users are still required to write Neo4j queries if conducting secondary development. To facilitate user adoption, particularly for integration with AI agents, we will encapsulate various KG retrieval and invocation functionalities into Command Line Interfaces (CLI). Additionally, for downstream tasks, we will distill the best practices identified during our experimental process into agentic skills, enabling one-click loading when utilizing agents for automated scientific discovery.

Integrating More Knowledge Forms. Currently, the scientific knowledge in our KG primarily encompasses papers, keywords, authors, and other paper-centric entities. However, the complete research workflow extends beyond these elements to include atomic knowledge, theorems and standards, experimental experiences, datasets and code, among others. How to acquire such knowledge and establish its associations with papers to form a more extensive and well-organized knowledge network that facilitates agentic utilization and reasoning constitutes a crucial research direction for our future work. **We argue that KG is an indispensable knowledge organization form for scientific discovery** because, although LLMs have achieved remarkable advancements in semantic understanding, they still exhibit substantial deficiencies in capturing logical relationships among knowledge entities, a capability of paramount importance for scientific research that transcends mere semantic associations.

Benchmark and Evaluation. Benchmarks serve as a critical engine driving scientific progress. Although automated scientific research has gained considerable popularity, numerous stages within this domain still lack high-quality benchmarks and evaluation metrics that faithfully simulate real-world research scenarios. Furthermore, many scientific tasks involve long-form responses, and the evaluation of such outputs is often ambiguous, making it difficult to establish definitive verifiers. KGs, as symbolic knowledge repositories, can provide essential reference points for such verification processes. Additionally, the knowledge stored within KGs can serve as valuable data sources for benchmark construction. In this paper, we merely present running examples of downstream tasks, remaining at the qualitative analysis level. In future work, we will develop dedicated benchmarks based on SCIATLAS to quantitatively assess the downstream application capabilities of agent scientists.

Dynamic Update. Currently, our KG updates primarily rely on periodic manual execution of fixed scripts. Although we support user-initiated updates, automated real-time updates are essential to keep pace with the rapidly evolving knowledge landscape. In future work, we will systematize the real-time update strategies mentioned in §2.3 to support daily KG update mechanisms.

6. Related Work

6.1. Automated Scientific Research

Recent breakthroughs in LLMs [Chen et al., 2025b, Qiao et al., 2023] have propelled them into a central position within the domain of Automated Scientific Discovery [Lu et al., 2024, Nie et al., 2026, Qiao et al., 2025, Schmidgall et al., 2025, Xie et al., 2025]. The complete workflow of automated scientific discovery comprises five consecutive phases: *i) Literature Reviewing*, during which LLMs search for papers on designated topics across the internet or specialized collections and consolidate them into organized surveys [Agarwal et al., 2025, Asai et al., 2024, Liang et al., 2025, Wang et al., 2024b, Yan et al., 2025]; *ii) Hypothesis Generation*, where LLMs leverage both their inherent parametric knowledge and acquired external information to formulate feasible research concepts

[Baek et al., 2025, Li et al., 2024, Su et al., 2025, Wang et al., 2024a, Zhao et al., 2025]; iii) *Method Implementation*, wherein LLMs convert the generated hypotheses into functional code, verify them via rigorous experimentation, and conduct statistical evaluation [Jiang et al., 2025, Liu et al., 2025, Novikov et al., 2025, Ou et al., 2025, Yu et al., 2025]; iv) *Manuscript Writing*, in which LLMs document the research rationale, technical approach, and experimental outcomes in the form of academic papers or reports [Chen et al., 2025a, Wen et al., 2024]; and v) *Peer Reviewing*, where LLMs assume the responsibilities of reviewers to assess manuscripts from multiple perspectives [Gao et al., 2024, Jin et al., 2024, Weng et al., 2025, Zhu et al., 2025]. The entire workflow of automated scientific discovery is an extremely knowledge-intensive process, in which literature review serves as the primary source of external knowledge beyond model parametric knowledge. Consequently, a precise scientific search is of paramount importance for the whole workflow.

6.2. Scientific Search and Discovery

Human scientists typically conduct scientific retrieval through general-purpose academic search platforms such as Google Scholar and Semantic Scholar, domain-specific preprint servers including arXiv, ChemRxiv, and PubMed, or official publisher platforms for journals and conferences. In the domain of automated scientific research, early efforts primarily relied on keyword or vector-based retrieval within local paper collections [Baek et al., 2025, Su et al., 2025, Wang et al., 2024a, 2025]. With the agentic advancement of LLMs, web-based literature resources have become accessible through API calling [Li et al., 2024, Qiao et al., 2026, Si et al., 2025, Tang et al., 2025, Zhang et al., 2025, Zhao et al., 2025]. Deep research agent frameworks can further leverage the semantic understanding and reasoning capabilities of LLMs to enable in-depth literature retrieval [Ju et al., 2025, Qian et al., 2026, Zhang et al., 2026]. However, these approaches not only incur high computational costs and response latency but also, due to the absence of deterministic cognitive maps as anchors for LLMs, render them highly susceptible to logical hallucinations within complex exploratory trajectories. So we argue that KG is an indispensable knowledge organization form for scientific discovery because, although LLMs have achieved remarkable advancements in semantic understanding, they still exhibit substantial deficiencies in capturing logical relationships among knowledge entities, a capability of paramount importance for scientific research that transcends mere semantic associations. A recent related work, OmniScientist [Shao et al., 2025], has also proposed a research knowledge base. However, it lacks the integration of core keywords for paper interconnection and semantic vectors. Furthermore, its Elasticsearch-based search algorithm merely relies on simple propagation through citation and reference relationships, without performing structured traversal and deep topological reasoning over heterogeneous subgraphs to uncover potentially relevant literature.

7. Conclusion

In this report, we introduce SCIATLAS, a large-scale, multi-disciplinary, heterogeneous academic knowledge graph designed as a panoramic scientific evolution network. By integrating 9 categories of entity nodes, 12 categories of relational edges, and over 43M papers, SCIATLAS provides a structured topological cognitive substrate that dismantles disciplinary barriers and furnishes AI agents with a global perspective. Furthermore, we develop a neuro-symbolic retrieval algorithm featuring tri-path collaborative recall and graph reranking, achieving a seamless transition from simple semantic matching to deterministic association discovery. We also present key application directions of SCIATLAS, including automated research trend synthesis, idea positioning, and academic trajectory exploration, to demonstrate that SCIATLAS can serve as an effective “cognitive map” to empower the full loop of automated scientific research while reducing reasoning costs.

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A. Full Schema of SCIATLAS

Table 3 | Node types and attributes in the Neo4j schema.

Node Type	Attribute	Type
Author	id	ID
Author	label	string
Author	display_name	string
Author	orcid	string
Author	works_count	int
Author	cited_by_count	int
Author	h_index	int
Author	i10_index	int
Author	mean_citedness_2y	float
Author	created_date	string
Author	updated_date	string
Domain	id	ID
Domain	label	string
Domain	display_name	string
Domain	description	string
Domain	works_count	int
Domain	cited_by_count	int
Domain	created_date	string
Domain	updated_date	string
Field	id	ID
Field	label	string
Field	display_name	string
Field	description	string
Field	works_count	int
Field	cited_by_count	int
Field	created_date	string
Field	updated_date	string
Institution	id	ID
Institution	label	string
Institution	display_name	string
Institution	ror	string
Institution	country_code	string
Institution	country	string
Institution	city	string
Institution	type	string
Institution	works_count	int
Institution	cited_by_count	int
Institution	h_index	int
Institution	homepage_url	string
Institution	created_date	string
Institution	updated_date	string
Keyword	id	ID
Keyword	label	string

Node Type	Attribute	Type
Keyword	text	string
Keyword	text_normalized	string
Keyword	frequency	int
Keyword	text_embedding	float[]
Paper	created_date	string
Paper	updated_date	string
Paper	pdf_url	string
Paper	pdf_source_id	string
Paper	pdf_source_display_name	string
Paper	pdf_source_type	string
Paper	pdf_is_oa	boolean
Paper	pdf_is_published	boolean
Paper	pdf_version	string
Paper	venue_source_id	string
Paper	venue_source_display_name	string
Paper	venue_source_type	string
Paper	venue_raw_source_name	string
Source	id	ID
Source	label	string
Source	display_name	string
Source	type	string
Source	issn_l	string
Source	is_oa	boolean
Source	is_core	boolean
Source	works_count	int
Source	cited_by_count	int
Source	created_date	string
Source	updated_date	string
Subfield	id	ID
Subfield	label	string
Subfield	display_name	string
Subfield	description	string
Subfield	works_count	int
Subfield	cited_by_count	int
Subfield	created_date	string
Subfield	updated_date	string
Topic	id	ID
Topic	label	string
Topic	display_name	string
Topic	description	string
Topic	keywords	string[]
Topic	works_count	int
Topic	cited_by_count	int
Topic	domain_id	string
Topic	field_id	string
Topic	subfield_id	string
Topic	created_date	string

Node Type	Attribute	Type
Topic	updated_date	string

Table 4 | Relationship types in the Neo4j schema.

Type	Source	Target	Properties
AFFILIATED_WITH	Author	Institution	is_current (boolean)
AUTHORED	Author	Paper	position (int), is_corresponding (boolean), raw_name (string)
CITES	Paper	Paper	none
COAUTHOR	Author	Author	count (int)
COOCCUR	Keyword	Keyword	count (int)
DOMAIN_OF	Field	Domain	none
FIELD_OF	Subfield	Field	none
HAS_KEYWORD	Paper	Keyword	relevance_score (float)
HAS_TOPIC	Paper	Topic	score (float), is_primary (boolean)
RELATED_TO	Paper	Paper	none
SUBFIELD_OF	Topic	Subfield	none

Table 5 | Indexes in the Neo4j schema.

Index Name	Type	Definition
paper_title_normalized_idx	RANGE	:Paper(title_normalized)
paper_text_ft	FULLTEXT	:Paper(title, abstract)
paper_title_ft	FULLTEXT	:Paper(title)
paper_abstract_ft	FULLTEXT	:Paper(abstract)
keyword_text_ft	FULLTEXT	:Keyword(text, text_normalized)

Table 6 | Vector indexes in the Neo4j schema.

Index Name	Node	Configuration
paper_title_embedding_idx	Paper	dimensions=1024, similarity=COSINE
paper_abstract_embedding_idx	Paper	dimensions=1024, similarity=COSINE
keyword_text_embedding_idx	Keyword	dimensions=1024, similarity=cosine

B. Prompts Used in this Report

B.1. Keyword Extraction

Keyword Extraction

System Prompt: You are an expert assistant that extracts high-level academic keywords for knowledge graph construction.

User Prompt: **Goal:** From an academic paper abstract, extract a small set of canonical, high-level keywords that represent the main research topics, tasks, or method categories of the paper.

These keywords will be used as entities in a knowledge graph to connect related papers across many scientific and engineering fields.

Important: The goal is NOT to capture detailed phrases from the abstract. The goal is to identify general, reusable concepts that are likely to appear in many different papers.

Requirements:

- Extract only 3-8 keywords.
- Prefer high-level research areas, problem types, method families, or evaluation paradigms.
- Normalize detailed wording into broader concepts when possible.
- Keywords should be reusable across many papers and suitable as knowledge graph entities.
- Use concise noun phrases (typically 1-4 words).
- If the abstract does not contain many strong high-level concepts, return fewer keywords.
- Also score each keyword's relevance to the abstract on a 1-10 integer scale.

Avoid:

- Long descriptive phrases copied from the abstract
- Paper-specific terminology or system names
- Highly customized or marketing-style expressions
- Implementation details or narrow technical descriptions

Good keyword examples (general, reusable concepts):

- machine learning
- wireless communication
- computer vision
- protein structure prediction
- finite element method
- Monte Carlo simulation
- graph neural networks
- energy optimization
- fault detection
- numerical simulation

Bad keyword examples (paper-specific phrasing):

- hierarchical dual-path adaptive learning framework
- multi-stage cross-modal feature fusion architecture
- lightweight energy-aware dynamic routing mechanism
- novel high-performance prototype system
- end-to-end task-specific optimization pipeline

Abstract: {abstract}

Output format: Return ONLY a JSON object:

```
{"keywords": ["keyword1", "keyword2"], "scores": [8, 7]}.
```

scores must be integers from 1 to 10, aligned with keywords, with no extra text.

B.2. Downstream Tasks

B.2.1. Idea Grounding – Query Generation

Idea Grounding – Query Generation: Generate dense-retrieval queries from structured scientific extraction results.

System Prompt: You are a scientific grounding agent. Generate retrieval-oriented paragraph search queries from structured scientific extraction results. Return strict JSON only.

User Prompt: Generate dense-retrieval queries from the structured extraction below.

Requirements:

- Use only the motivation and method sections as query sources.
- Consider all provided sentences from those two sections before deciding which queries to emit.
- Select the most retrieval-useful items yourself. Do not mirror every input sentence if some are redundant.
- Produce at most max_queries total queries across both sections combined.
- You may allocate the total freely across motivation and method.
- Each output item must contain:
 - section: either motivation or method
 - sentence: the source sentence you selected
 - query: the final retrieval-oriented rewrite
- Keep the selected sentence meaning exactly.
- Write concise academic retrieval phrases or sentences likely to match paper paragraphs.
- Preserve task, object, method, mechanism, training objective, dataset, baseline, metric, or analysis anchors when present.
- Avoid vague wording such as "the framework", "this method", "evaluation", "performance improvement", or "how it works".
- Do not introduce unsupported facts.

Return JSON with this schema only:

```
{
  "items": [
    {
      "section": "motivation | method",
      "sentence": "selected source sentence",
      "query": "retrieval-oriented rewrite optimized for paragraph retrieval"
    }
  ]
}
```

Structured extraction: {items_json}

B.2.2. Idea Grounding – Grounding

Idea Grounding – Grounding: Analyze how retrieved evidence aligns with a research idea unit.

System Prompt: You are a scientific grounding alignment agent. Given a research idea unit, a retrieved paragraph, and local paper context, analyze how the retrieved evidence aligns with the target idea. Use only the provided evidence. Return strict JSON only.

User Prompt: Analyze how the retrieved evidence aligns with a specific research idea unit.

Research idea: {idea_text}

Target query section: {query_section}

Target query sentence: {query_sentence}

Retrieval query: {query_text}

Paper title: {paper_title}

Paper abstract: {paper_abstract}

Section path: {section_path_text}

Retrieved paragraph: {paragraph_text}

Previous paragraphs: {previous_paragraphs}

Next paragraphs: {next_paragraphs}

Requirements:

- Judge whether the retrieved evidence supports the target query unit.
- Use the paragraph as the primary evidence and use the surrounding context only to clarify scope, subject, or omitted details.
- Do not invent claims that are not grounded in the provided paragraph or context.
- If the paragraph is weakly related, say so explicitly.
- `focus_aspect` should state which specific aspect of the target query is actually addressed by the evidence.
- `grounded_passage` should be a concise evidence-focused passage, usually 1-3 sentences, that is more context-aware and better aligned to the target query than the raw paragraph alone.
- `evidence_span` should quote or closely paraphrase the most directly relevant span from the retrieved paragraph or immediate context.
- `shared_points` should list concrete aspects that are aligned between the target idea unit and the evidence.
- `different_points` should list concrete mismatches, missing parts, narrower scope, or different emphasis between the evidence and the target idea unit.
- `coverage_label` must be one of: `high`, `partial`, `limited`, `none`.
 - Use `high` only when the evidence covers most of the target idea unit.
 - Use `partial` when there is clear overlap but also important missing coverage.
 - Use `limited` when the evidence is only weakly related or covers a small sub-aspect.
 - Use `none` when it is essentially irrelevant.
- `why_this_matches` should briefly explain the overall judgment.

Return JSON with this schema only:

```
{
  "status": "supported | partially_supported | weak_match | irrelevant",
  "focus_aspect": "which specific aspect of the target query is grounded",
  "grounded_passage": "context-aware grounding passage",
  "evidence_span": "most relevant evidence span",
  "shared_points": [...],
  "different_points": [...],
  "coverage_label": "high | partial | limited | none",
  "why_this_matches": "brief explanation of the match quality"
}
```

B.2.3. Idea Generation

Idea Generation: Propose novel research ideas from a collection of papers.

System Prompt: You are an expert research idea generator and only return valid JSON.

User Prompt: You are an expert research idea generator. Given the papers below, propose novel research ideas that extend, combine, or contrast this work. Return strict JSON only with this schema:

```
{
  "ideas": [
    {
      "title": "concise idea title",
      "description": "2-3 sentence core description of the idea",
      "novelty": "why this is novel compared to existing work",
      "significance": "potential impact and importance",
      "key_references": ["paper title 1", "paper title 2"]
    }
  ]
}
```

Generate exactly {idea_count} ideas.

Papers:

{paper_context}

B.2.4. Research Trend Predicting

Research Trend Predicting: Summarize research trends from chronologically ordered papers.

System Prompt: You analyze academic topic evolution and only return valid JSON objects.

User Prompt: You are an expert research trend analyst. Given chronologically ordered papers from one topic, summarize the research trend. Return strict JSON only with this schema:

```
{
  "one_sentence_summary": "...",
  "trend_summary": "...",
  "stage_summary": [
    {"period": "...", "theme": "...", "description": "..."}
  ],
  "methodological_shifts": ["..."],
  "emerging_topics": ["..."],
  "open_gaps": ["..."],
  "future_directions": ["..."],
  "representative_papers": [
    {"title": "...", "year": 2024, "why_representative": "..."}
  ]
}
```

Papers: {papers_by_year}

B.2.5. Author Research Profile

Author Research Profile: Summarize a researcher's major directions from their publication list.

System Prompt: You summarize an author's research trajectory and only return valid JSON objects.

User Prompt: You are an expert academic intelligence analyst. Summarize one researcher's major directions from their publication list. Return strict JSON only with this schema:

```
{
  "author_name": "{author\_name}",
  "overall_academic_profile": "...",
  "main_research_directions": [
    {"theme": "...", "active_years": "...", "description": "..."}
  ],
  "technical_arsenal": ["..."],
  "representative_papers": [
    {"title": "...", "year": 2024, "why_representative": "..."}
  ]
}
```

Papers: {paper_lines}